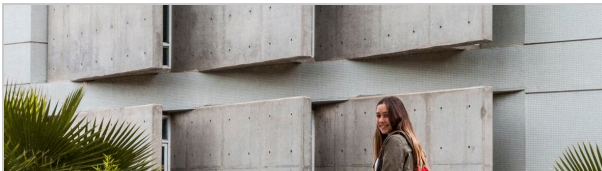


1



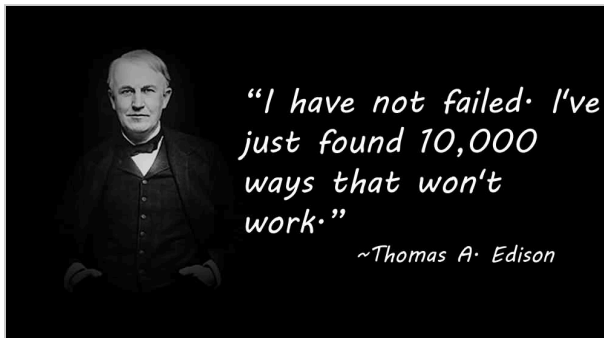
Datacommunicatie en netwerken

Bachelor in de elektromechanica – automatisering

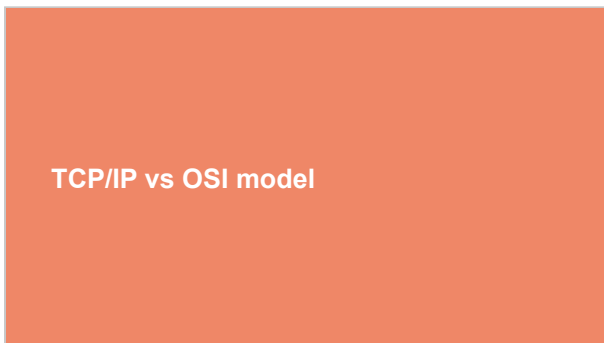
Tom Demets | Stefan Lievens

tom.demets@hogent.be | stefan.lievens@hogent.be

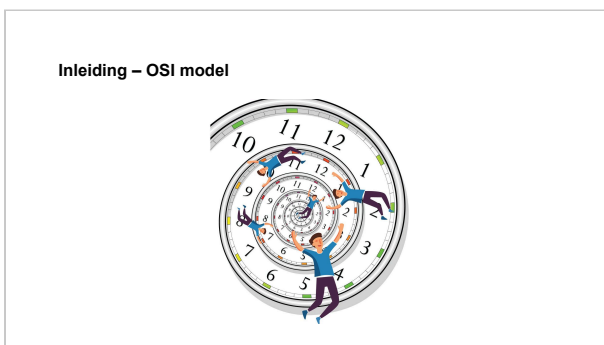
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4



Inleiding – OSI model

5

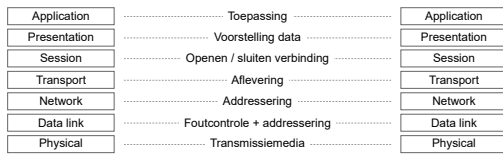
Inleiding – OSI model

Wat zou er nodig zijn qua hardware, afspraken, ... om wereldwijd computers met elkaar te laten communiceren?

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Inleiding – OSI model

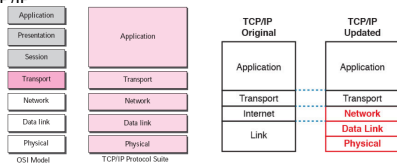
- OSI = Open Systems Interconnection (1979)
 - Set van afspraken ontwikkeld om communicatie tussen verschillende hardware mogelijk te maken => lagenmodel



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Inleiding – TCP/IP model

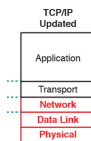
- Vereenvoudigd OSI model
 - Applicatie laag neemt Application, Presentation en Session over
- OSI in de praktijk, praktisch ieder netwerkapparaat ondersteunt TCP/IP



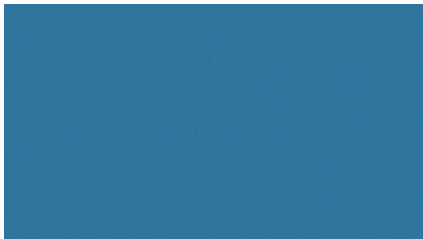
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Inleiding – TCP/IP model

- De werking van een computernetwerk begrijpen
- =
- Het TCP/IP model begrijpen
- We bespreken systematisch de verschillende lagen
- Please Do Not Take Anything (TCP/IP)*
- Please Do Not Take Something Precious Away (OSI)*



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<https://www.youtube.com/watch?v=cA9ZJdqzOoU>

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TCP/IP model / Laag 1: fysieke laag

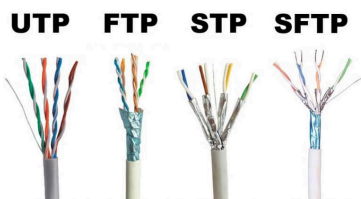
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Ethernetkabel

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Ethernetkabel – connectoren

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Ethernetkabel – afscherming

14

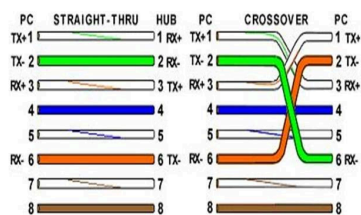
Ethernetkabel – categorie

Categorie	+/- 30m	+/- 50m	100m	Connector
cat5e				
cat6				
cat6a				
cat7				
cat8				

Hoe kan je zien als een kabel cat5e, cat6, cat6a, cat7 of cat8 is? / Wat als je verder moet dan 100m? / Wat in de industrie?

Oefening: ga op onderzoek en vul de tabel aan.

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Ethernetkabel – straight-thru en crossover

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Oefening RJ45 vs M12



M12 versus RJ45 Ethernet connection systems

Dietmar Rörig, Global Key Account Manager
Phoenix Contact GmbH & Co. KG, Blomberg, Germany

Abstract

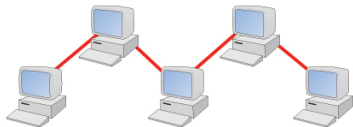
When it comes to connectivity in a facility plant environment, the people would agree upon the superiority of the M12 connector system over the standard RJ45 connector and cables, a connector originally designed to do no more than hook up a telephone extension. Industrial Ethernet calling requires connection solutions that are sturdy and reliable.

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M12 vs RJ45	3

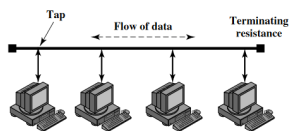
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Line topologie



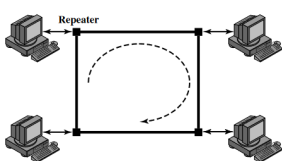
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Bus topologie



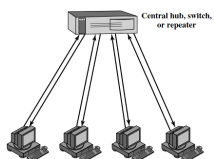
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Ring topologie



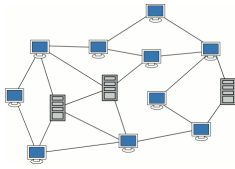
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Ster topologie



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Mesh topologie



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